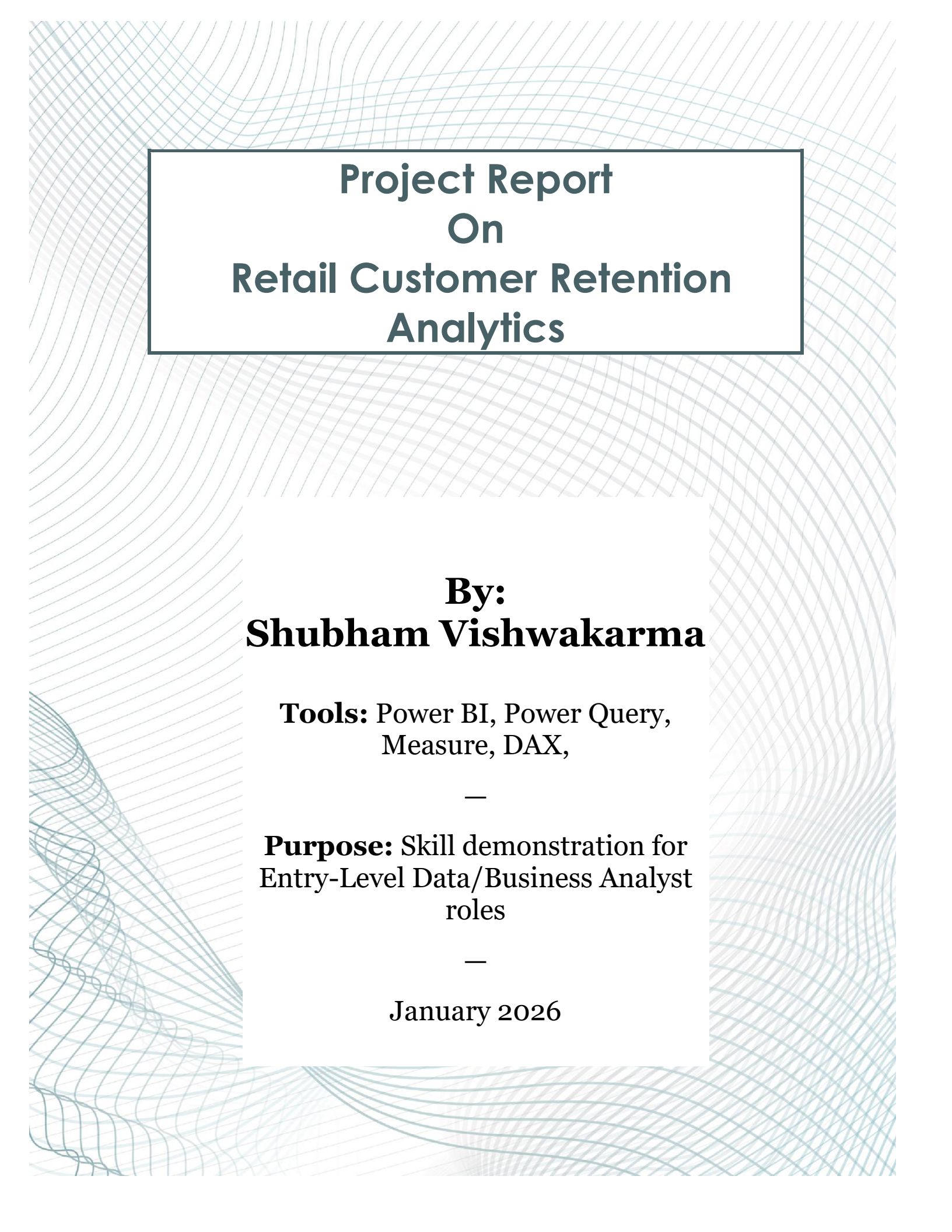




Project Report On Retail Customer Retention Analytics

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Tools: Power BI, Power Query,
Measure, DAX,

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Purpose: Skill demonstration for
Entry-Level Data/Business Analyst
roles

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Executive Summary

Retail company operates in a highly competitive retail environment with customers engaging through superstores, express outlets, and online channels. Despite having rich datasets covering customer demographics, transactions, loyalty programs, and store operations, existing reporting lacks the analytical depth required to proactively manage customer retention. Increasing churn and uneven loyalty engagement have created a need for a centralized, insight-driven analytics solution.

This project delivers an interactive Customer Retention Analytics Dashboard using Power BI, integrating multiple data sources to provide a unified view of customer behaviour, churn risk, loyalty effectiveness, and store-level performance. The primary objective is to enable company to identify why customers churn, recognize high-value and at-risk customers, and design targeted, data-backed retention strategies.

Project Objective

The primary objective of this project is to design and deliver a comprehensive, interactive Customer Retention Analytics solution using Power BI that enables retail company to better understand customer behaviour, reduce churn, and maximize customer lifetime value across regions, store formats, and loyalty tiers.

This project aims to transform siloed operational data into a single, trusted analytical view by integrating customer demographics, transaction history, store attributes, loyalty program data, and churn indicators. By doing so, the dashboard supports both strategic and operational decision-making for senior leadership, regional managers, and marketing teams.

Key Objectives

- **Enable Data-Driven Retention Decisions**

Identify churn patterns and retention risks across customer segments, regions, income groups, and store types, allowing TESCO to move from reactive reporting to proactive retention management.

- **Understand Customer Value and Loyalty Behaviour**

Analyse repeat purchase behaviour, loyalty tier effectiveness, and promotion usage to distinguish high-value customers from low-engagement and at-risk segments.

- **Evaluate Loyalty Program and Promotion Impact**

Measure how loyalty tiers, points accumulation, redemption behaviour, and promotional offers influence customer retention, purchase frequency, and spending patterns.

- Link Store Performance to Retention Outcomes**
 Assess how store characteristics such as location, store type, and operational maturity affect transaction value and customer churn, enabling store-specific optimization strategies.
- Prioritize High-Value Customers Using CLV**
 Calculate and segment Customer Lifetime Value (CLV) to identify customers who contribute most to long-term revenue and those requiring targeted reactivation or retention efforts.
- Deliver Actionable Insights Through an Interactive Dashboard**
 Provide dynamic filtering and drill-down capabilities across region, income group, loyalty tier, and store type, empowering stakeholders to quickly translate insights into action.

Data Description

Customer Demographics:

Column Name	Data Type	Description
Customer_ID	Text	Unique identifier for each customer
Gender	Text	Gender of the customer (Male/Female)
Age	Whole Number	Customer's age in years
Membership_Sin	Date	Date the customer joined TESCO's loyalty/membership program
Marital_Status	Text	Marital status (Single, Married, Divorced)
Region	Text	Geographic region assigned to the customer
Income_Group	Text	Income classification (Low, Medium, High)

Customer Transactions:

Column Name	Data Type	Description
Transaction_ID	Text	Unique transaction ID
Customer_ID	Text	ID of the customer who made the transaction
Transaction_Date	Date	Date of the transaction
Store_ID	Text	Store where the transaction occurred
Product_Category	Text	Product category purchased (Grocery, Electronics, etc.)
Amount	Decimal	Total monetary value of the transaction
Quantity	Whole Number	Number of units purchased
Promotion_Applied	Text	Whether a promotion was applied (Yes/No)

Store Locations:

Column Name	Data Type	Description
Store_ID	Text	Unique identifier for each store
City	Text	City where the store is located
Region	Text	Region of the store (aligned with customer regions)
Store_Type	Text	Type of store (Superstore, Express)
Opening_Year	Whole Number	Year the store was opened

Loyalty Program:

Column Name	Data Type	Description
Customer_ID	Text	ID of the customer enrolled in the program
Loyalty_Tier	Text	Loyalty tier (Silver, Gold, Platinum)
Points_Earned	Whole Number	Total loyalty points earned
Points_Redeemed	Whole Number	Points used/redeemed by the customer
Last_Redemption_Date	Date	Date of most recent points redemption

Churn Labelled Customers:

Column Name	Data Type	Description
Customer_ID	Text	Unique ID of the customer
Last_Transaction_Date	Date	Date of the most recent transaction
Churned (Yes/No)	Text	Churn label: "Yes" if the customer is inactive > 180 days
Days_Since_Last_Purchase	Whole Number	Number of days since last transaction (as of analysis date)

Tasks Performed

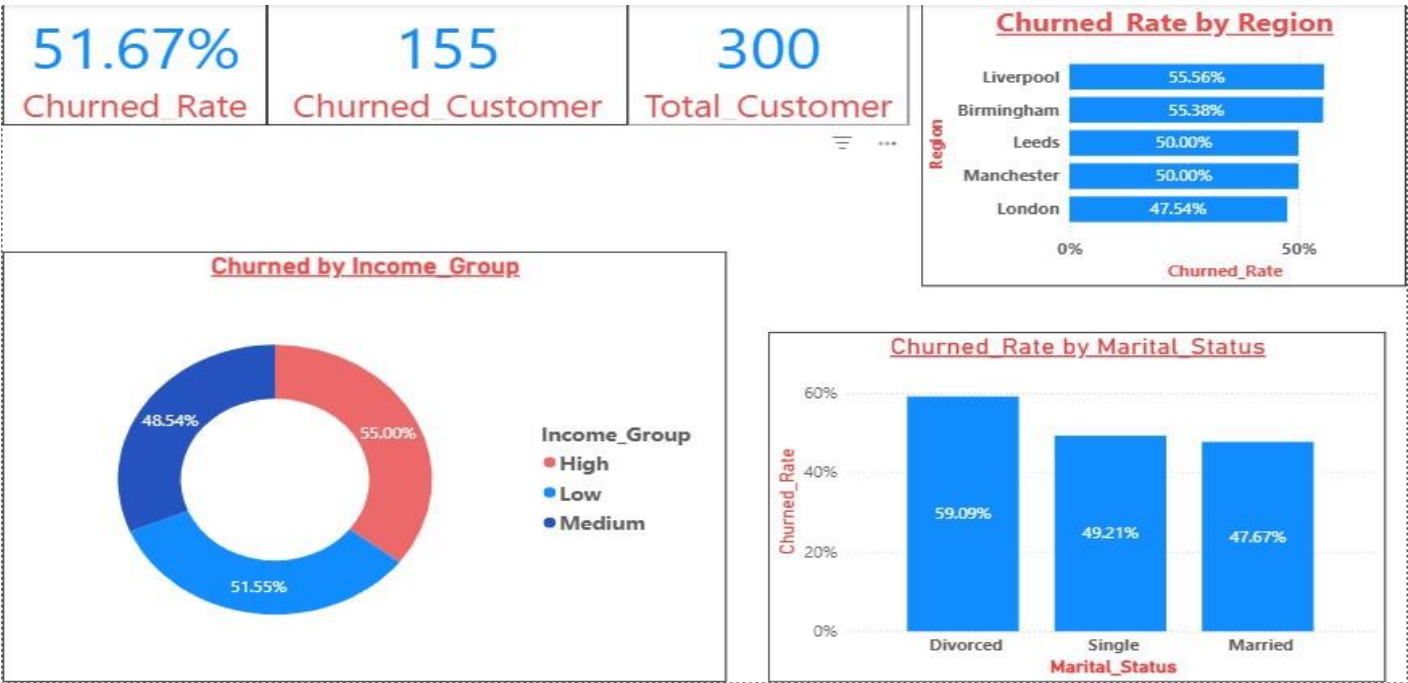
Data Modelling and Cleaning:

The analysis begins with data preparation and modeling, where all datasets are cleaned, standardized, and integrated using Power BI's Power Query Editor. Key transformations include removal of duplicates, handling missing values, formatting of date and numeric fields, and creation of derived attributes such as membership duration, transaction year, and transaction month. A robust relational data model is then established to ensure

accurate cross-table analysis across customers, transactions, loyalty programs, store locations, and churn indicators.

Churn and Retention Metrics:

Once the data foundation is established, the project focuses on customer retention and churn analysis. Churn rates are calculated and analyzed across multiple dimensions including region, income group, store type, and loyalty tier. Customer funnels are used to visualize the transition from total customers to repeat buyers and churned customers, helping identify critical drop-off points in the customer lifecycle.



Repeat Purchase Analysis:

Repeat purchase behavior is analyzed using DAX-based measures to calculate transaction frequency at the customer level. Customers are dynamically segmented into low, mid, and high purchase-frequency tiers, enabling real-time loyalty classification through slicers. Average purchase frequency is compared across regions, age groups, and loyalty tiers to identify strong and weak engagement patterns. This analysis helps highlight loyal customer segments as well as early indicators of disengagement for targeted retention actions.



Promotion & Loyalty Impact:

Promotional effectiveness is analyzed by comparing transaction behavior with and without promotions using conditional DAX measures. Loyalty program performance is evaluated through churn comparison across loyalty tiers and points earned versus redeemed analysis, highlighting opportunities to improve customer engagement and retention.



Store Performance vs Retention:

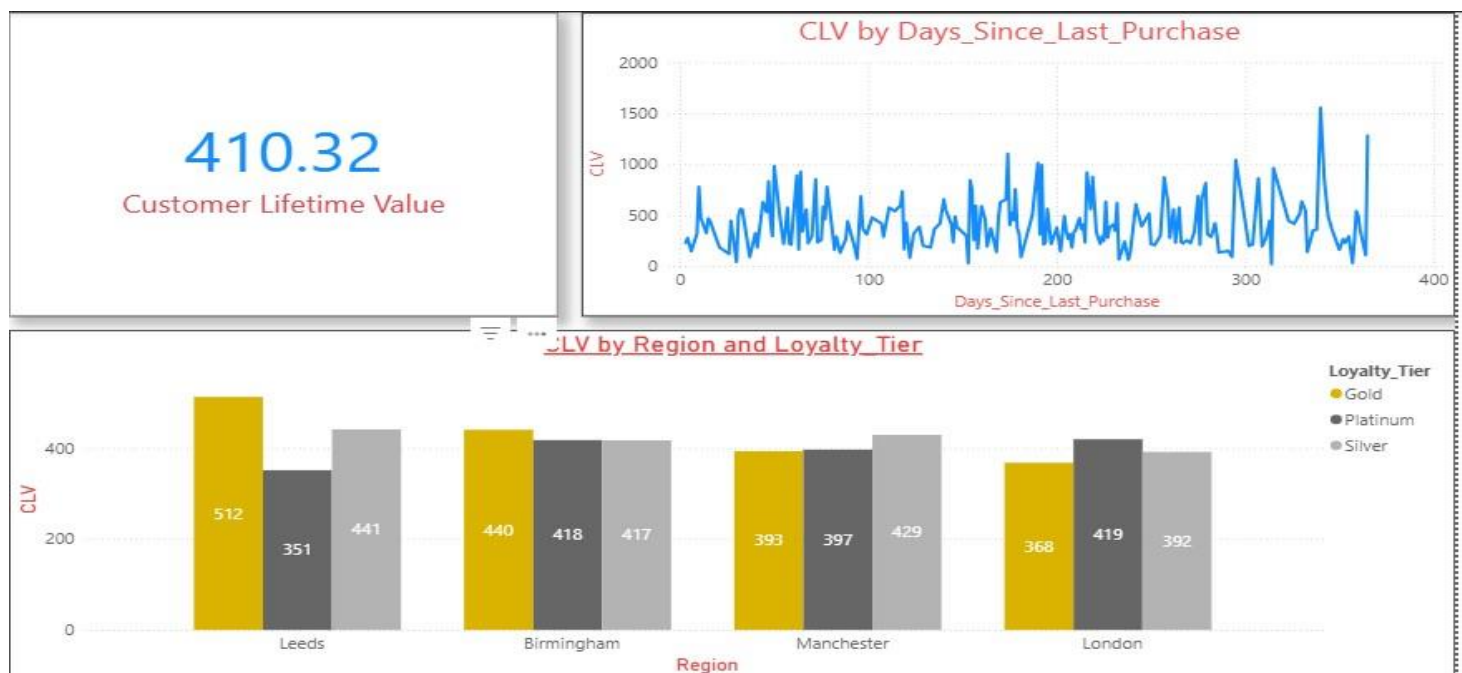
Store-level analysis examines average transaction value and churn rates across store types and regions. The relationship between store maturity and customer retention is analyzed

to identify underperforming locations and guide store-specific marketing and retention initiatives.



Customer Value (CLV) Analysis:

Customer Lifetime Value is calculated using total spend and membership duration to segment customers into low, medium, and high-value groups. CLV is analyzed against recency and loyalty tiers to identify high-value at-risk customers and prioritize retention and reactivation efforts.



Recommendations

1. Make focused campaign based on store type & region for retention of customers. Example: - “Win Back Customer” for churned customer.
2. Make proper strategies for redemption of earned points as points earned are not redeemed. Example: - “With certain points earned customer can buy products of their liking or can reduce the bill amount by points redemption.
3. Can focus on Loyalty Tier & Income group customer by offering special discount or early access to festive deals.
 - Tesco should focus on retaining the customer by analysing the past store data and based on results create campaigns specific for region, age group, income group etc, to engage more customer.
 - Next Tesco should target adults age group, Divorced group, the loyalty tier as their churn rate is high, find the reason and resolve the issues. Example: - “Look if the competing store is offering better discounts or there is an issue with products by getting the feedback from customers”.